Lohith Burra

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EU Chancenkarte visa - authorized to work in Germany/EU immediately, no sponsorship required

SUMMARY

AI and Automation Engineer who turns manual, judgment-heavy processes into reliable tools other people can run without him. Deep Python: packages, generators, HTTP and CLI surfaces, deterministic validation layers and test harnesses, built inside stacks chosen by someone else rather than rewritten. Mechanical engineering undergraduate, MS in Data Science. EU Chancenkarte visa, authorised to work in the EU immediately, no sponsorship required.

CORE SKILLS

Python & Packaging: Python 3.11, package and framework design, FastAPI, http.server with ThreadingHTTPServer, argparse, JSON Schema, pytest, unittest, custom test runners, Docker

AI & Agents: Claude API, Gemini, LangGraph, MCP tools, RAG, pgvector, embeddings, structured output and schema constraint, LLM benchmarking and cost measurement

Automation & Orchestration: n8n (23+ workflows, 2 production systems), workflow generation from Python, GitHub Actions CI, REST API integration, webhooks, Airtable API, Google Drive API

Data & Platform: Apache Spark, PySpark, ETL pipelines, AWS (S3, Lambda, Glue), PostgreSQL, MySQL, TypeScript, Next.js, Vercel

PROFESSIONAL EXPERIENCE

Manager, AI & Automation Jun 2026 - Sep 2026

Corominas Consulting - Remote, Germany

- Built a config-driven multi-tenant engine in Python that turned a manual judgment-heavy process into an unattended pipeline: 8 stages exposed as both a CLI and an HTTP surface, per-instance behaviour carried in configuration rather than code branches, state written atomically under a lock
- Wrote the deterministic validation layer that makes generated output safe to ship: checks in 10 families written against the artefact rather than the generator, so swapping the model does not invalidate them, backed by 715 assertion calls across 33 test scripts. Graded 9 models over 40 human-labelled items and moved production onto one costing $0.0109 against a $1.149 reference
- Treated orchestration as generated infrastructure rather than hand-built console state: four Python generators emit and maintain 26 workflow definitions totalling 490 nodes, reading templates and logic from the real source modules at build time so orchestrator and package cannot drift, with a validator refusing any artefact whose connections do not hold
- Built the enablement layer, not just the code: reusable per-instance scaffolding, a preflight readiness script naming each missing item mechanically, and contract tests that fail when a rule is traded away

AI Automation Developer Aug 2025 - Jun 2026

Formula Dream - Remote | Contract

- Built an end-to-end unattended pipeline: self-scheduling Python render jobs on GitHub Actions writing outputs under a filename contract, feeding a 21-node orchestration with dual triggers, an event trigger plus an hourly sweep, so nothing is missed and nothing fails silently
- Made retries safe by establishing idempotency first: outputs marked with custom properties for dedup, every service-facing HTTP node retrying 3 times with backoff, failures emailing the exact node and execution link
- Built a meta-layer on top of the orchestrator: a generator emitting complete importable workflow JSON pre-configured for the target credentials, idea to running workflow in under 10 minutes

AI Engineer Apr 2025 - Mar 2026

Think Round Inc - Remote, US

- Built a production conversational system in Python and TypeScript with memory persistence, confidence scoring and multi-layered safety controls over LLM output, on a multi-agent architecture supporting parallel development across three use cases on one shared foundation

Program Analyst Feb 2022 - Aug 2023

Cognizant Technology Solutions - India

- Built automated ingestion pipelines in Python and Apache Spark processing 50M+ rows daily, cutting data-to-KPI latency from 48 hours to 2
- Optimised ETL workflows on AWS (S3, Lambda, Glue), measurably reducing infrastructure cost

KEY PROJECTS

SceneAgent - Input-agnostic 3D scene intelligence pipeline Apr 2026

- Video, photos, point cloud, BIM or a pre-trained splat in, a semantically queryable scene out: segmentation, backprojection and DBSCAN clustering detecting 151 objects, with a LangGraph agent exposing scene operations as MCP tools and the scene queried from the backend rather than carried in context. FastAPI, Next.js, pgvector, Kubernetes. Built in 48 hours

EDUCATION

MS Data Science, Stevens Institute of Technology, USA (2025) - BTech Mechanical Engineering, JNTU K, India (2022)